

FUD-1

Combustion chamber design review

250 lbf

Nitrous oxide and ethanol-water

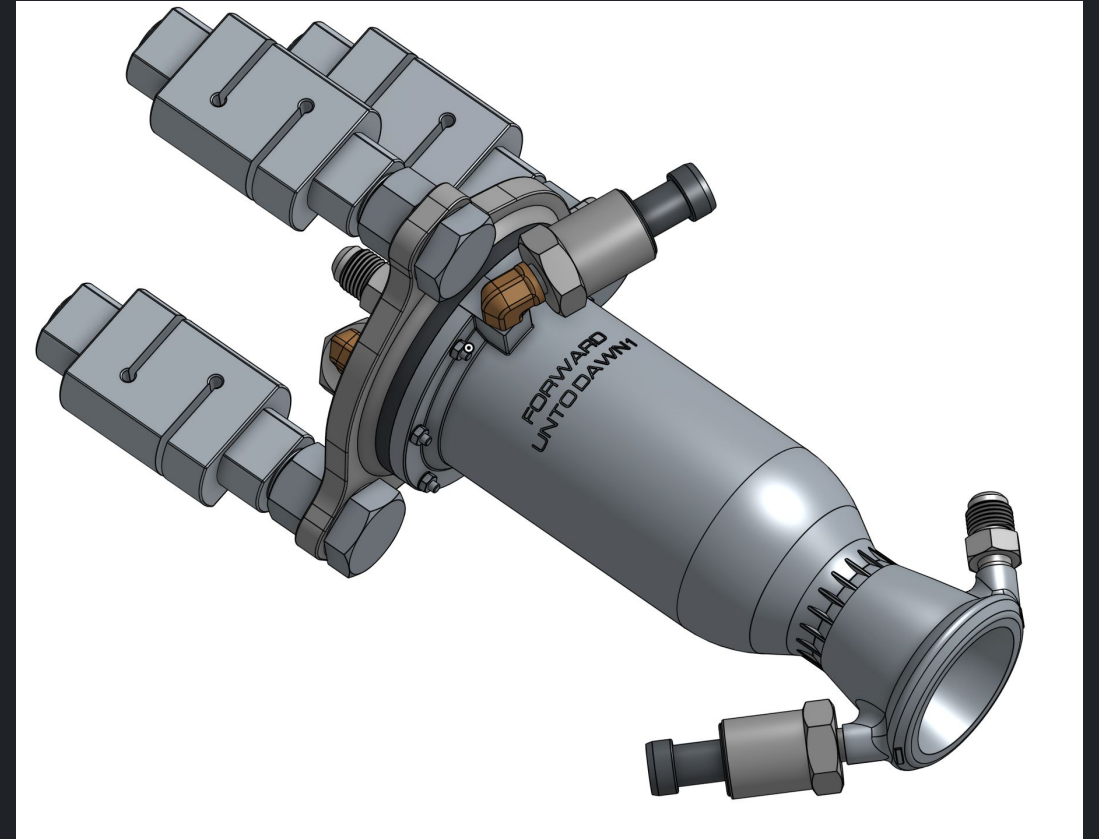
Regeneratively cooled, printed 316L

2 second first burn

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After ordering the print

Ordering the chamber fixes the geometry. Everything in the right column can still change afterwards.

FIXED

- Channel cross-section and height
- Inner and outer wall thickness
- Chamber and nozzle contour
- Chamber length and L^*
- Transducer and igniter boss locations
- Manifold routing and port positions
- Build orientation

STILL ADJUSTABLE

- Fuel blend ratio, currently 90/10
- O/F ratio and chamber pressure
- Film cooling fraction
- Injector pressure drop and annulus gap
- Pintle geometry, a separate machined part
- Injector plate and flange
- Burn duration, sequencing and test plan

Requirements and locked parameters

250

lbf thrust, sea level

200

psia chamber pressure

3.0

O/F ratio

2

second first burn

PARAMETER	VALUE	DRIVER
Oxidizer	Nitrous oxide, self-pressurizing, 920 psia at 30 C	No pressurant needed on the ox side
Fuel	90 / 10 ethanol and water by mass	Lower flame temperature at no lsp cost
Cooling	Regenerative, 38 axial channels, plus 2% film	Reusable article, multiple firings
Chamber material	SS316L, laser powder bed fusion	Cost and print availability
Injector	Ox-centered pintle	Simpler fuel routing, easier to machine
First burn	2 seconds, then longer burns if it looks good	Validate the hardware before committing to a long run
Article type	Ground test only, not flight hardware	No mass budget, so thermal margin is cheap

PROCESS

Design process

Geometry was frozen before any CAD was drawn.

- 1 NASA CEA**
Fuel blend and O/F selection
- 2 Sizing spreadsheet**
Throat, chamber and nozzle geometry
- 3 RPA verification**
Checked my spreadsheet, reconciled against CEA
- 4 Geometry freeze**
Contour and L^* locked
- 5 Regen cooling**
Axial versus helical, channel sizing
- 6 Injector**
Pintle sizing and annulus
- 7 Structures**
Pressure vessel and bolted joints
- 8 CAD**
Onshape, started after freeze
- 9 Manufacturing**
Print planning and post-machining
- 10 Test plan**
Cold flow, stand, sequencing, abort logic

Propellant selection

Adding water lowered temperature and molecular weight together

	PURE ETHANOL	90 / 10
Combustion temp	2945 K	2868 K
Exhaust MW	about 25	23.1

Isp depends on temperature divided by molecular weight, so the two effects largely cancel. Isp held while wall temperature dropped. The V-2 used the same trade at 75/25.

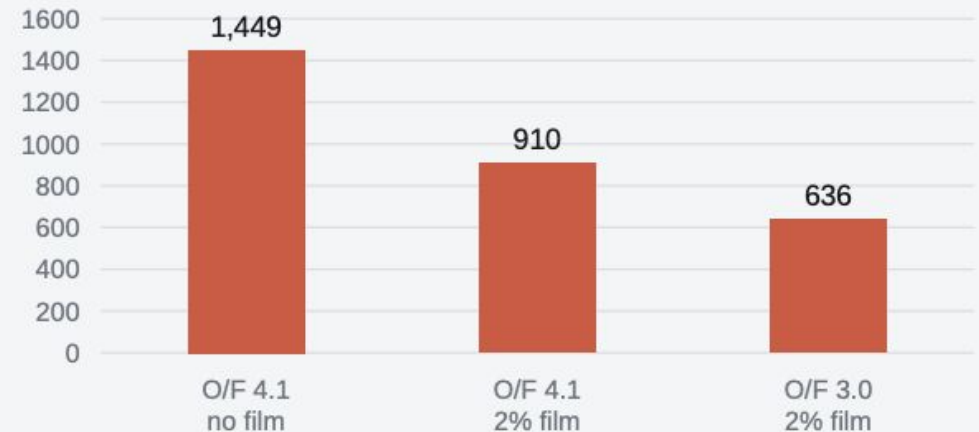
Why the blend stops at 10% water

- Past roughly 15 to 20 percent water the cancellation stops holding and performance drops for real.
- Water is the coolant knob. I turned it as far as the thermal problem required and stopped.
- It is also adjustable after the print, since it is what goes in the tank.

O/F 3.0 instead of the peak-Isp 4.2

Peak Isp sits at O/F 4.2 to 4.4. At O/F 4.1 the throat wall came out at 1449 K with regen cooling, which 316L does not survive. O/F 3.0 gives more fuel for cooling and a lower flame temperature.

That trades roughly 5% Isp for a chamber that survives the burn. With no mass budget the trade is cheap.



Geometry sized by hand first

Uses delivered Isp

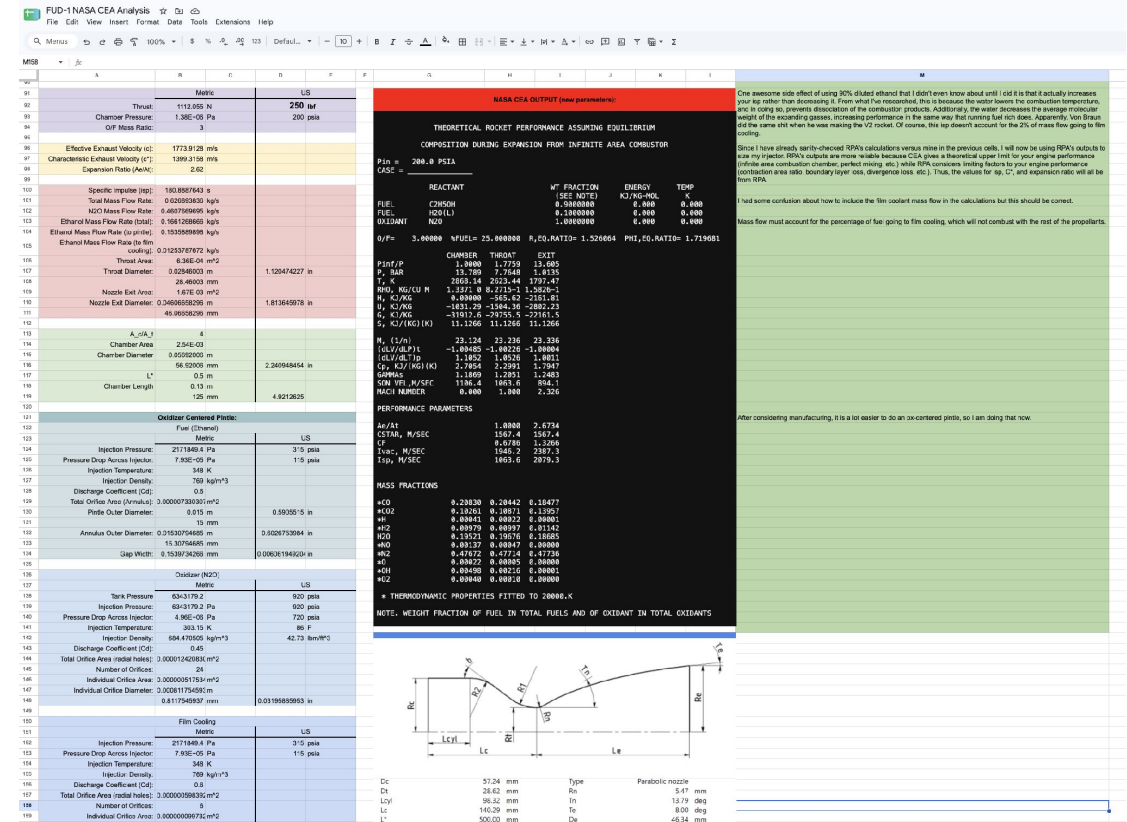
Pc is set by the throat

Contraction ratio

$$L^* = 500 \text{ mm}$$

Sea-level optimum

Sizing spreadsheet and the CEA case it was built from



Checking the sizing against RPA

RPA and CEA gave different performance numbers due to an assumption.

CEA assumes the combustion chamber is infinitely large. RPA lets you enter the real chamber size, and a real chamber performs slightly worse than an infinite one. So RPA gives lower numbers than CEA by design. Once I compared them on the same basis they agreed, and I used the RPA values because they account for the engine I am actually building.

Sizing uses delivered performance, not ideal

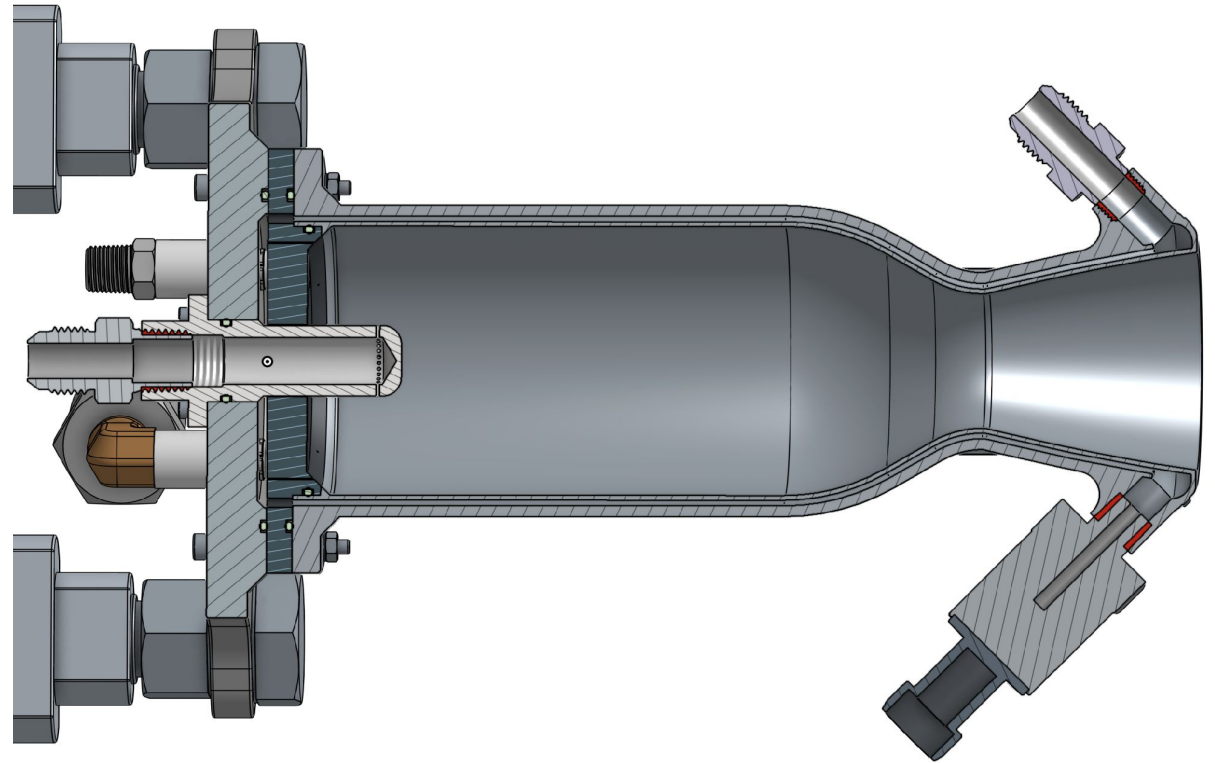
PARAMETER	IDEAL	DELIVERED, USED FOR SIZING
Specific impulse	195.4 s	180.9 s
Characteristic velocity c*	1567.4 m/s from CEA	1399.3 m/s
Effective exhaust velocity		1773.9 m/s
Reaction efficiency		0.9675
Nozzle efficiency		0.9787
Overall efficiency		0.9469

The efficiency factors are RPA defaults rather than measurements. Measured performance on the first fire is the check on them.

Frozen geometry

CHAMBER AND NOZZLE

Chamber diameter D_c	57.24 mm
Throat diameter D_t	28.62 mm
Exit diameter D_e	46.34 mm
Cylindrical length	98.32 mm
Total chamber length	140.29 mm
Characteristic length L^*	500 mm
Contraction ratio A_c/A_t	4.00
Expansion ratio A_e/A_t	2.62
Nozzle type	Parabolic bell
Nozzle length L_e	45.34 mm
Initial and exit angle	13.79 and 8.00 deg
Converging half-angle	30 deg



Section view showing the full contour, regen channels and injector

The CAD chamber is about 3 mm longer than the RPA contour for a more generous L^* . A 45 degree chamfer runs where the chamber wall meets the injector plate.

Regenerative cooling iteration

Every cooling decision was made against maximum gas-side wall temperature at the throat.

CONFIGURATION	MAX Twg	OUTCOME
No film cooling, O/F 4.1	1449 K	Rejected, 316L does not survive
2% film, O/F 4.1	910 K	Marginal
O/F 3.0, helical channels at 30 deg	625 K	Good, but unprintable near the throat
O/F 3.0, axial, 1.5 mm channel height	654 K	Baseline
O/F 3.0, axial, 1.2 mm channel height	636 K	Selected, matches the drawing
2 mm inner wall	860 K	Rejected
Conservative correlation, same geometry	750 K	Bounding case, still acceptable

Axial over helical

Helical was better by 30 K but drove wall thickness below 0.8 mm near the throat, which is not printable. Manufacturability broke a thermal tie.

Channel height

Reduced from 1.5 to 1.2 mm. A smaller channel makes the coolant move faster, which pulls more heat out of the wall. Channels are arc wedges following the wall curvature, not rectangles.

Thermal results at final geometry

Thermal analysis

Location, mm	Radius, mm	Conv.heat coeff, kW/m²·K	Conv.heat flux, kW/m²	Rad.heat flux, kW/m²	Total heat flux, kW/m²	Twg, K	Twl, K	Twc, K	Tc, K	pc, MPa	wc, m/s
81.94	28.62	0.5062	574.4131	0.0000	574.4131	472.8837	472.8837	436.9677	324.0875	2.1946	2.9282
86.49	28.62	0.5062	574.4896	0.0000	574.4896	472.6680	472.6680	436.7473	323.0267	2.1951	2.9248
91.04	28.62	0.5061	574.5685	0.0000	574.5685	472.4463	472.4463	436.5207	321.9619	2.1956	2.9213
95.59	28.62	0.5061	574.6345	0.0000	574.6345	472.2573	472.2573	436.3277	320.8932	2.1961	2.9178
98.32	28.62	0.5061	574.6756	0.0000	574.6756	472.1402	472.1402	436.2081	320.2512	2.1964	2.9157
100.59	28.56	0.5081	576.8375	0.0000	576.8375	472.3846	472.3846	436.3174	319.7136	2.1966	2.9140
102.87	28.38	0.5143	583.3474	0.0000	583.3474	473.3312	473.3312	436.8573	319.1717	2.1968	2.9123
105.15	28.07	0.5249	594.4865	0.0000	594.4865	475.0224	475.0224	437.8526	318.6222	2.1971	2.9105
107.42	27.64	0.5405	610.7565	0.0000	610.7565	477.5305	477.5305	439.3443	318.0614	2.1973	2.9087
109.70	27.08	0.5618	632.9402	0.0000	632.9402	480.9667	480.9667	441.3946	317.4851	2.1976	2.9068
111.97	26.38	0.5902	662.1727	0.0000	662.1727	485.4865	485.4865	444.0880	316.8886	2.1978	2.9049
114.25	25.53	0.6274	700.2301	0.0000	700.2301	491.3429	491.3429	447.5666	316.2662	2.1981	2.9029
116.53	24.54	0.6762	749.5256	0.0000	749.5256	498.8513	498.8513	451.9950	315.6111	2.1984	2.9009
118.80	23.38	0.7409	813.8876	0.0000	813.8876	508.5266	508.5266	457.6488	314.9143	2.1986	2.8986
121.08	22.08	0.8255	896.5598	0.0000	896.5598	520.7431	520.7431	464.6995	314.1731	2.1989	2.8963
123.35	20.77	0.9270	993.4172	0.0000	993.4172	534.7825	534.7825	472.6866	313.3944	2.1992	2.8938
125.63	19.45	1.0489	1106.4429	0.0000	1106.4429	551.0111	551.0111	481.8521	312.5801	2.1995	2.8913
127.91	18.14	1.1969	1238.9049	0.0000	1238.9049	569.8365	569.8365	492.3998	311.7279	2.1998	2.8887
130.18	16.84	1.3758	1392.7197	0.0000	1392.7197	591.2905	591.2905	504.2415	310.8440	2.2001	2.8861
132.46	15.79	1.5489	1534.7226	0.0000	1534.7226	610.7567	610.7567	514.8334	309.9587	2.2003	2.8835
134.73	15.04	1.6878	1643.4951	0.0000	1643.4951	625.4084	625.4084	522.6874	309.0880	2.2006	2.8809
137.01	14.56	1.7778	1710.2598	0.0000	1710.2598	634.0962	634.0962	527.2027	308.2293	2.2008	2.8784
139.28	14.33	1.8078	1729.0654	0.0000	1729.0654	635.9793	635.9793	527.9106	307.3817	2.2011	2.8759
140.29	14.31	1.8007	1721.6019	0.0000	1721.6019	634.5695	634.5695	526.9673	307.0107	2.2012	2.8748
144.84	15.26	1.5038	1474.6096	0.0000	1474.6096	596.5814	596.5814	504.4155	305.3878	2.2017	2.8700
149.39	16.32	1.2822	1284.8223	0.0000	1284.8223	568.2140	568.2140	487.9089	303.8537	2.2022	2.8655
153.95	17.35	1.1164	1137.8144	0.0000	1137.8144	545.9341	545.9341	474.8161	302.4138	2.2027	2.8613
158.50	18.34	0.9889	1021.4515	0.0000	1021.4515	528.0969	528.0969	464.2507	301.0506	2.2033	2.8574
163.05	19.28	0.8911	929.9499	0.0000	929.9499	514.0129	514.0129	455.8848	299.7498	2.2038	2.8536
167.60	20.17	0.8039	847.5091	0.0000	847.5091	500.4691	500.4691	447.4926	298.5050	2.2043	2.8501
172.15	21.01	0.7365	782.5249	0.0000	782.5249	489.7545	489.7545	440.8387	297.3116	2.2048	2.8466
176.70	21.80	0.6809	728.2437	0.0000	728.2437	480.5723	480.5723	435.0484	296.1602	2.2053	2.8434
185.63	23.17	0.5981	646.3570	0.0000	646.3570	466.2225	466.2225	425.8154	294.0000	2.2063	2.8372

RPA axial thermal solution. Peak values occur at the throat, location 139 mm.

PEAK VALUES

Gas-side wall Twg	636 K
Coolant-side wall Twc	528 K
Total heat flux	1729 kW/m2
Coolant inlet	294 K
Coolant at 82 mm	324 K
Coolant pressure	2.19 MPa

316L holds up well here

636 K at the throat is far below where 316L loses strength, and the conservative case still comes in at 750 K.

One thing to check

The coolant-side wall reaches 528 K, which is above where ethanol boils at this pressure. There may be some boiling in the channels.

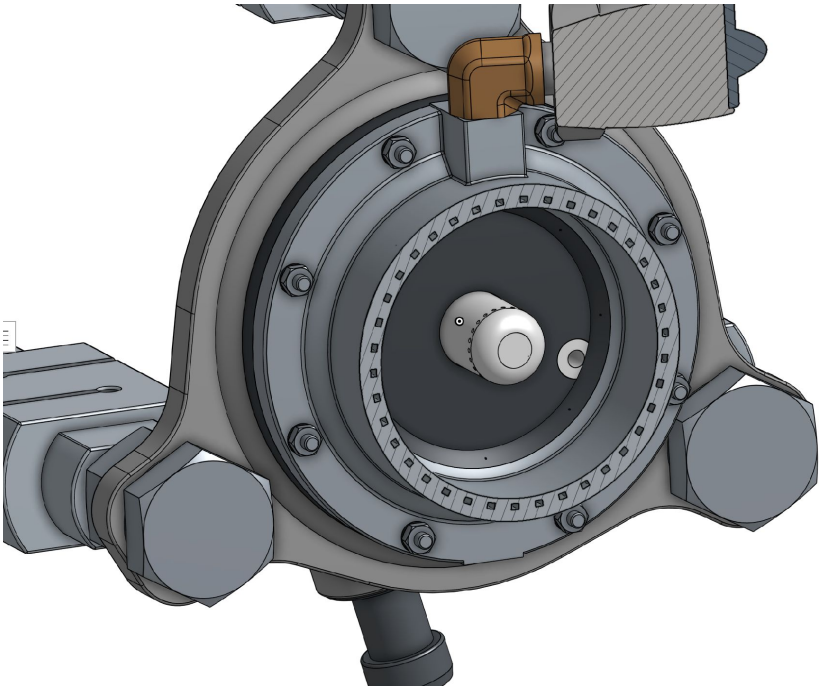
Injector, ox-centered pintle

OXIDIZER SIDE

Pintle outer diameter	15 mm
Pintle wall thickness	2 mm
Orifices	24 at 0.81 mm
Discharge coefficient	0.45
Injection pressure	920 psia
Pressure drop	720 psi

FUEL SIDE, ANNULUS

Injection pressure	315 psia
Pressure drop	115 psi
Injection temperature	348 K
Discharge coefficient	0.6
Gap at 15 mm pintle	0.154 mm
Film cooling holes	6 orifices



Section through the injector showing pintle, annulus and film cooling ring

OPEN

Fuel annulus gap

The gap is 0.154 mm, which is hard to hold concentric. Lowering the fuel pressure drop would widen it, but that moves toward the minimum drop needed for stable flow. If the gap ends up uneven, one side of the chamber runs hotter than the other.

Hoop stress and load paths

The channels make the wall too complicated to solve exactly by hand, so I bounded it. Each case removes a different piece of support.

CASE	WHAT IT ASSUMES	STRESS	YIELD	SF
Outer wall	Holds 300 psia coolant against atmosphere through 2 mm, near room temperature	32.9 MPa	170 MPa	5.2
Inner wall, worst case	Ribs and coolant ignored. A bare 1 mm tube holding 200 psia, hot	40.2 MPa	110 MPa	2.7
Inner wall, real net load	Coolant outside at 300 psia, chamber inside at 200, so 100 psi net inward	20.1 MPa	110 MPa	5.5
Full wall section	Inner and outer wall acting together through the ribs, 3 mm of metal	14.1 MPa	110 MPa	7.8

Governing number

The case that assumes the least still carries a safety factor of 2.7. Where the real stress falls between these bounds does not change any decision.

Thermal stress governs, and that is expected

The 108 K gradient across the inner wall gives about 240 MPa, so the hot face yields in compression. Normal for a regen chamber, and not a burst risk at 40 MPa hoop. It makes cyclic life the limit, not static strength.

STEP 9

Manufacturing plan

Printed

Chamber and nozzle in SS316L by laser powder bed fusion. The Xometry quote is about \$1,070 for the bare geometry.

Post-machined

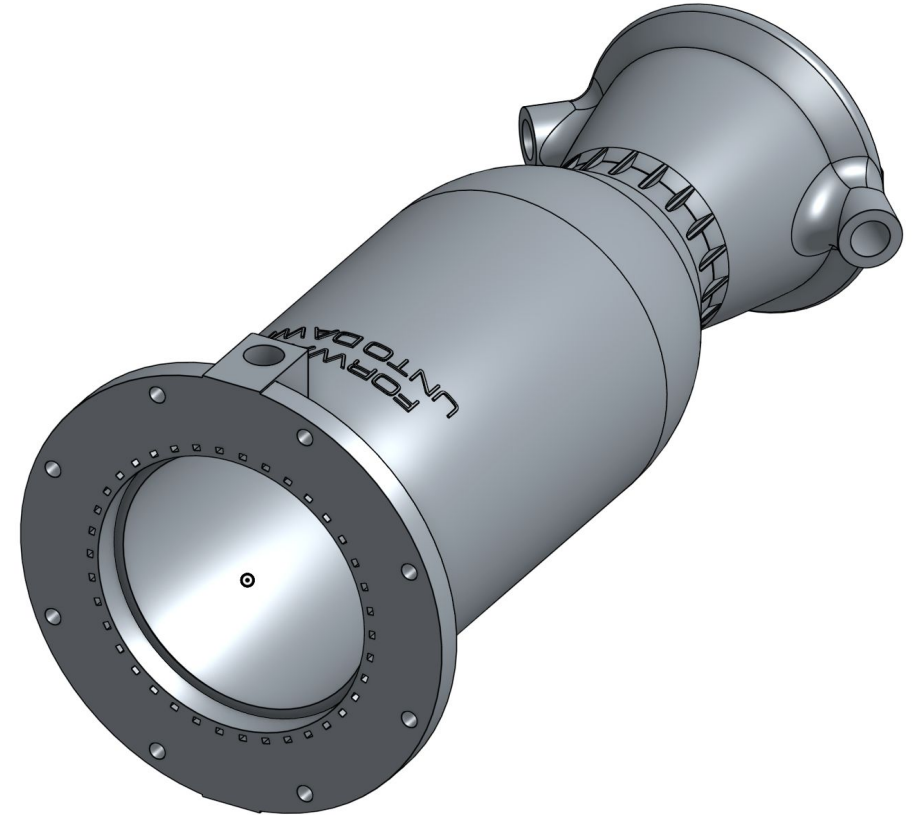
All threads. Printed threads add thousands to the quote, so bores print smooth and get tapped afterwards.

Machined separately

Pintle and igniter boss from SS303 bar. Injector plate and flange from half inch 6061 plate, used at stock thickness. Film cooling holes are in the injector plate, not the printed part.

Open with the vendor

Depowdering of the 38 channels. Protolabs will not guarantee internal passages.



The printed part: chamber, nozzle, channels, manifolds and flange in one piece

CAVEATS

What I am still uncertain about

Things worth knowing before you take any of these numbers at face value.

Efficiency factors are defaults

The performance efficiencies came from RPA defaults, not from anything I measured. Measured performance on the first fire is the real check on them.

The chamber is on the short side

L^* of 500 mm gives less residence time than is typical for this kind of propellant. A pintle mixes well, but if it is too short it shows up as lower than predicted performance.

Roughness is not in the model

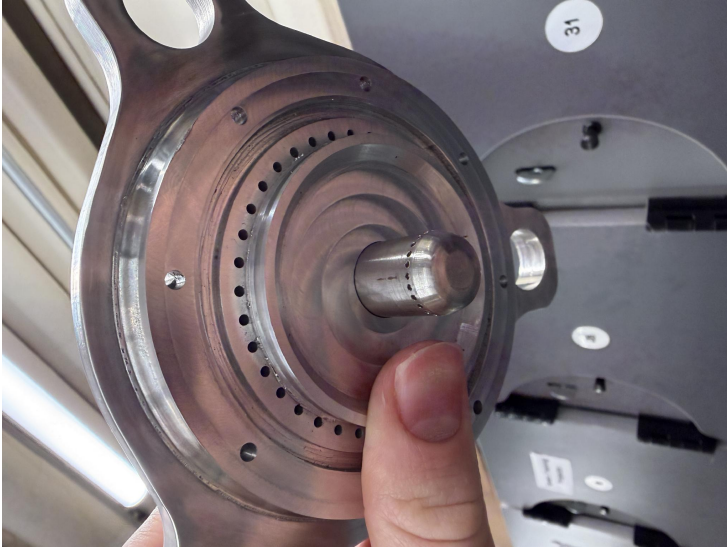
The thermal and pressure drop work assumes smooth channel walls. A printed channel is rough inside, which changes both the heat transfer and the pressure drop through the cooling circuit.

Plus the fuel annulus gap on the injector, which is the one I would most like a second opinion on.

WHERE THE BUILD IS NOW

Hardware status

Injector hardware is machined. The chamber print is not ordered yet.



Pintle fitted through both injector plates



Pintle, SS303, 24 oxidizer orifices



Injector plate and flange, 6061

Two known issues from machining. The 1/4 NPT nitrous port in the pintle was tapped off-axis, so it is being sealed with Loctite 567 and proof tested on its own to 1000 psi before it goes near the engine. The fit test also showed an uneven annulus gap, which will be corrected with three shims at 120 degrees and then dowelled in place.